Calibrating carbon sequestration with flux tower measurements

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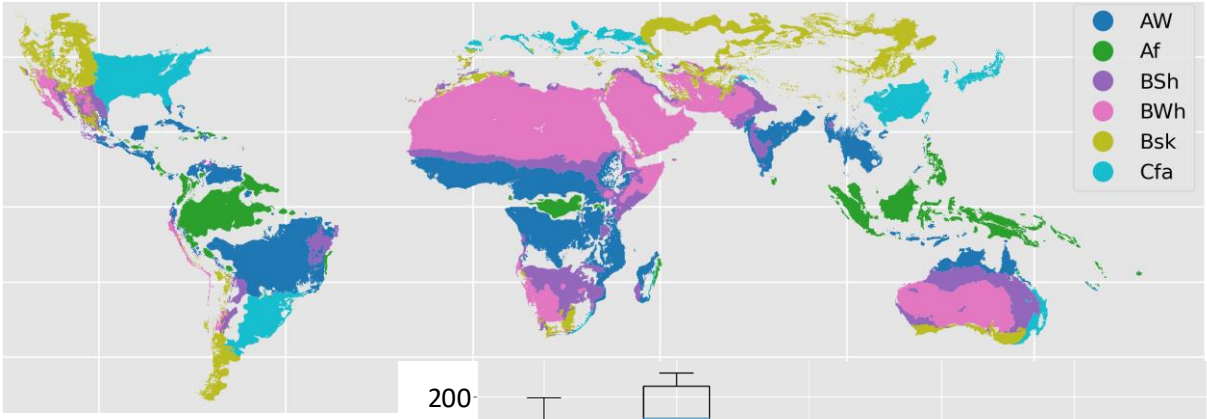
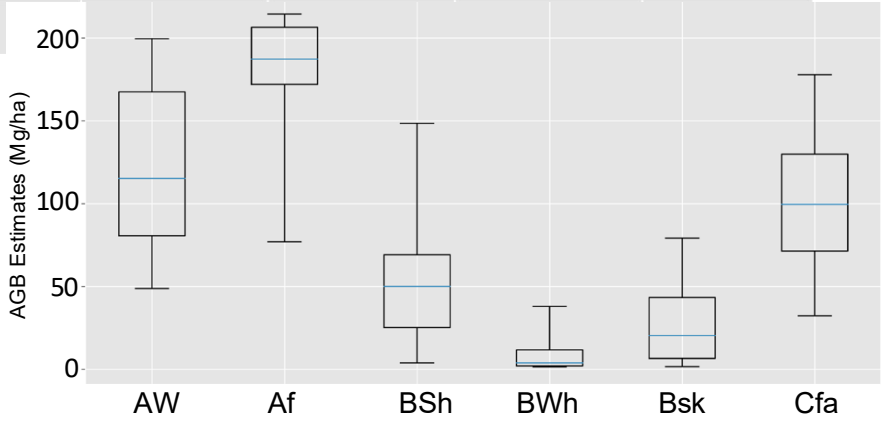
IBM Reserach

Why study Carbon Sequestration?

Maintaining and, ultimately, increasing **vegetation coverage** is likely the **most impactful** approach to **globally capture carbon**.

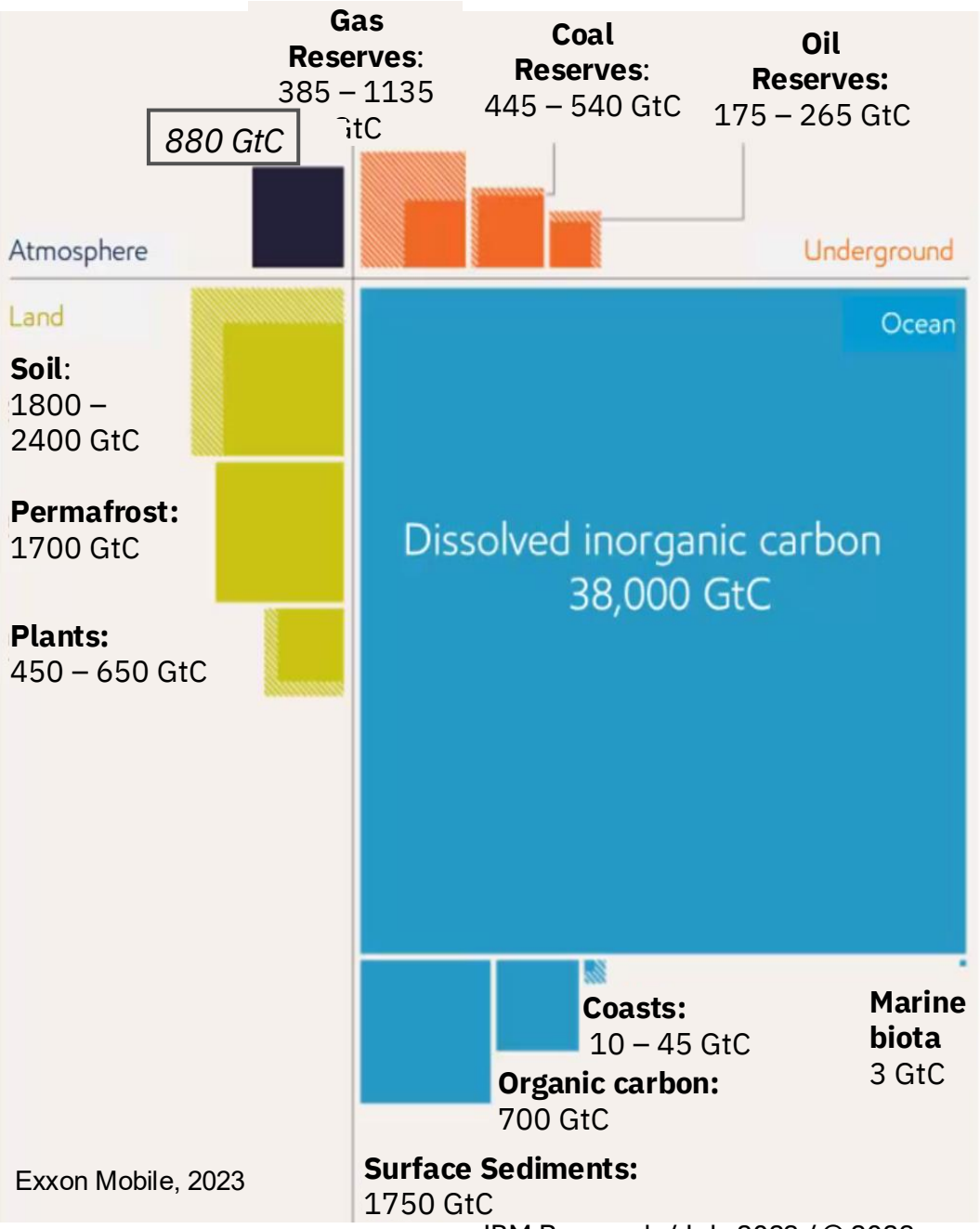
Biomass is a crucial **parameter** for **quantifying carbon** stored in vegetation, and **estimating it poses challenges**.





Biomes
AW: Tropical wet
Af: Tropical rainforest
Bsh and **Bsk**: Semiarid
BWh: Arid,
Cfa: Humid subtropical

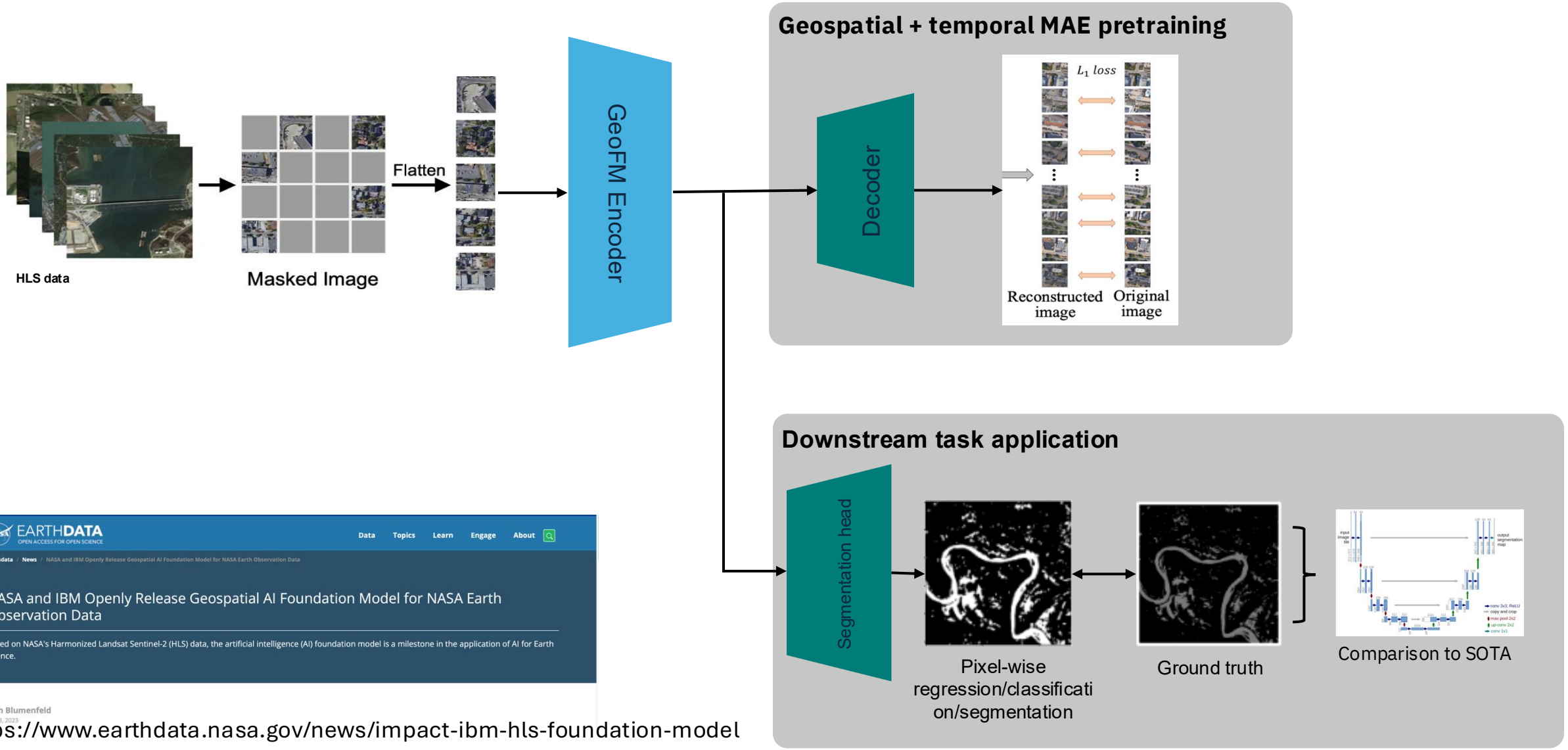
Nathaniel, J., Klein, L. J., Watson, C. D., Nyirjesy, G., & Albrecht, C. M. (2022). Aboveground carbon biomass estimate with Physics-informed deep network. *arXiv preprint arXiv:2210.13752*.



Exxon Mobile, 2023

IBM Research / July 2023 / © 2023 IBM Corporation

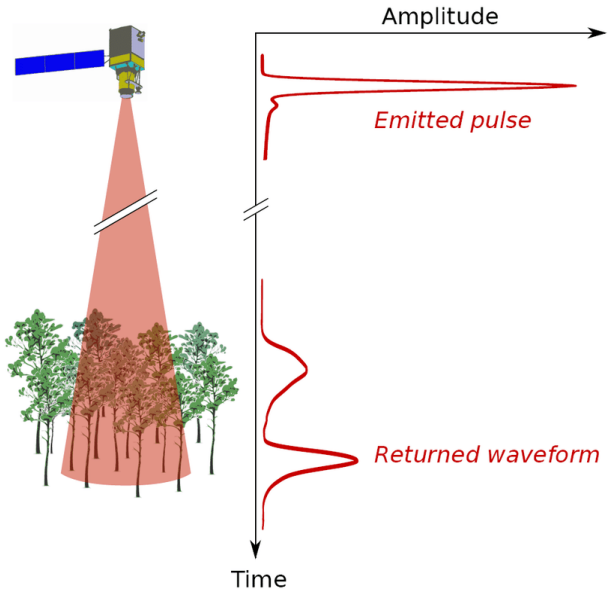
Geospatial Foundation Models



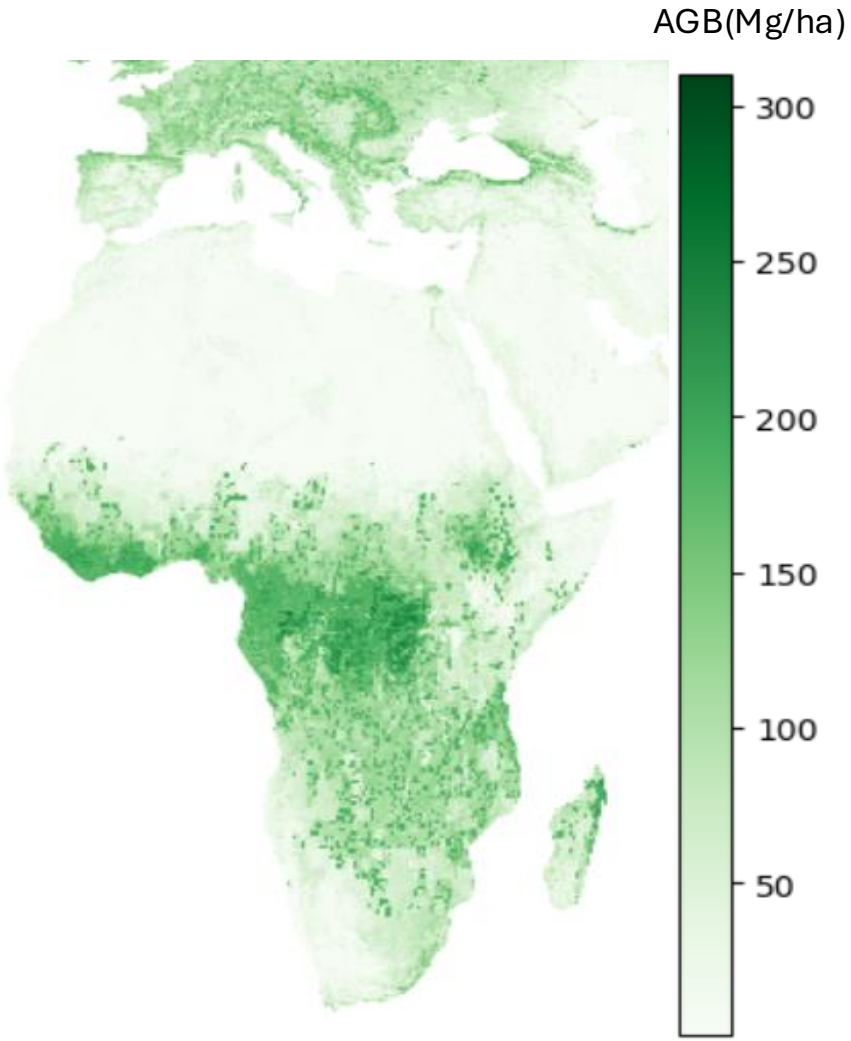
https://www.earthdata.nasa.gov/news/impact-ibm-hls-foundation-model

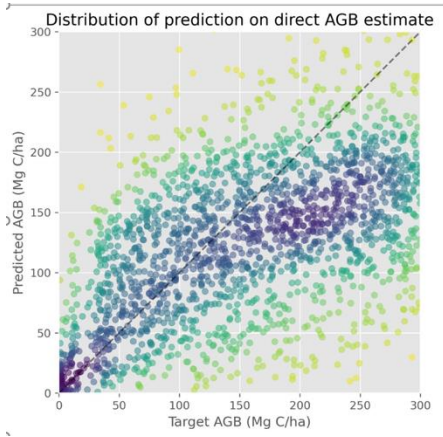
Biomass Estimation using Geo-Foundation Models

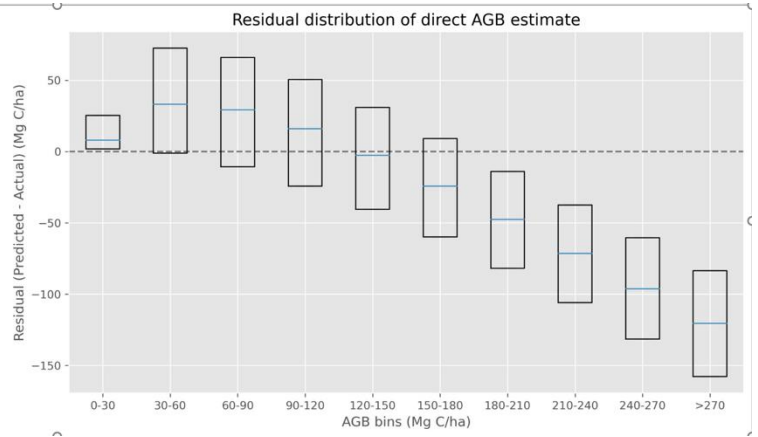
LIDAR Signal



Natural Carbon Stock Estimation







Predicting tree growth

Estimation	Prediction
Present	Future

- Central focus in ecology, forestry research
- Carbon sequestration dynamics
- Forest management strategies
- Climate change adaptation

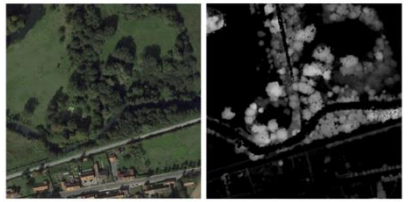
Existing model for predicting tree growth are sparse in time and space:

- Flux Tower measurement –sparse locations but very precise and covering many years
- Multiyear LiDAR measurements (NEON sites)-need to convert height data to carbon (allometric equations) hard to generalize
- Tree characteristic measurement (Diameter at Breast Height, Tree Height and using an allometric equations)

- **Physiological Principles** Predicting Growth (3-PG) since 1990s
- Physics based model simulating stand-level forest growth
- 100+ parameters, tabular based data

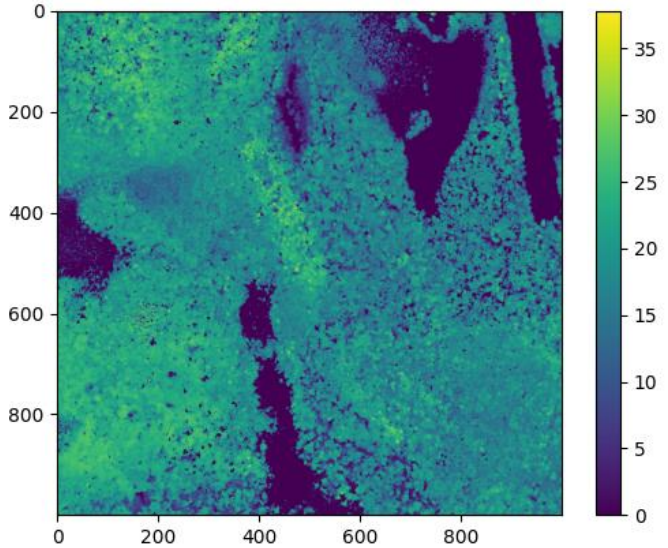
🚨**But… what if we want to train an AI-based tree-species agnostic height prediction model?**

Need relevant data => Very high resolution imagery + canopy height maps (CHMs) and Flux Tower to calibrate

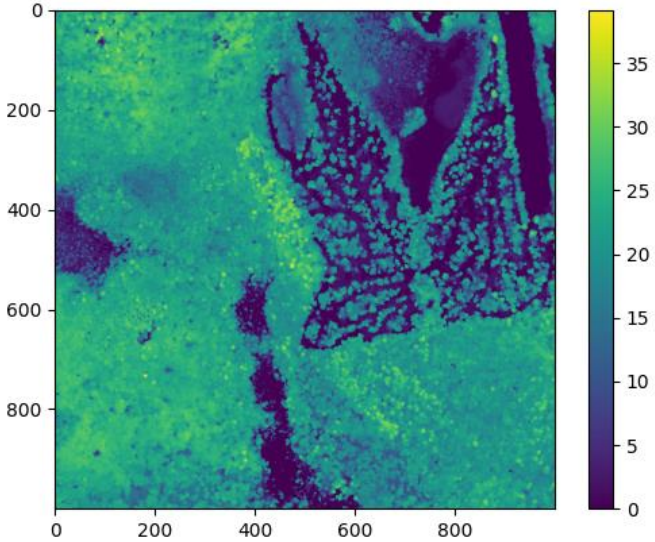


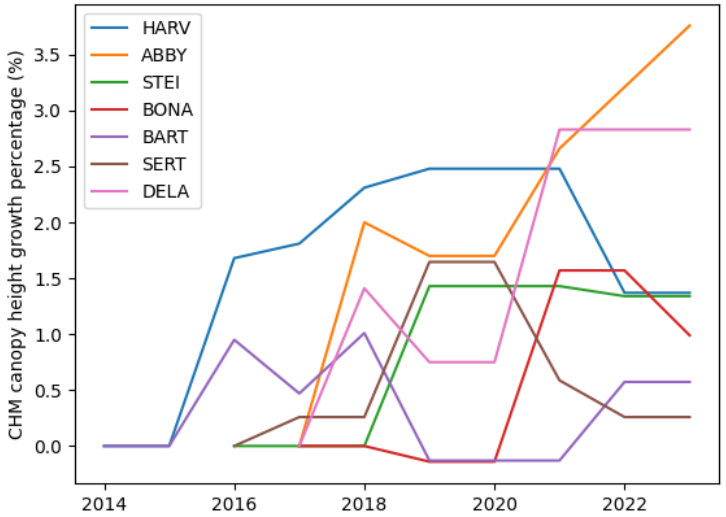
LiDAR measurement

2014



2022

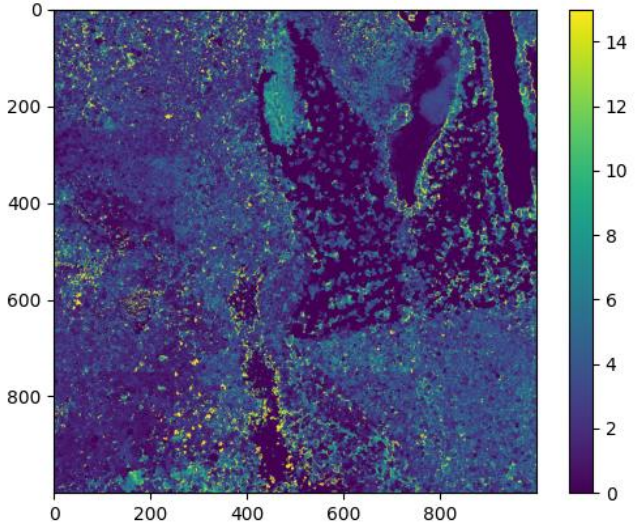




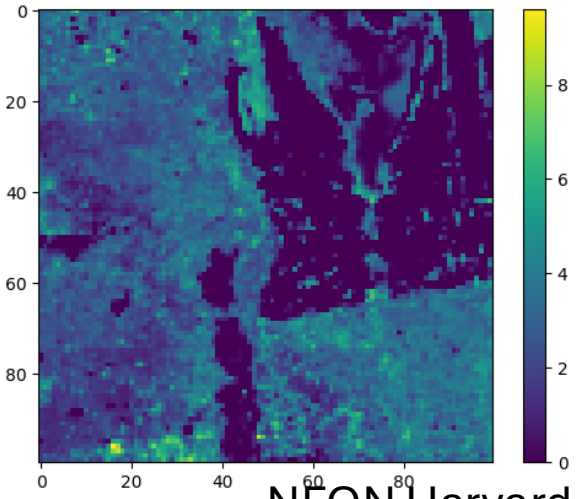
-LIDAR provides estimate of height with accuracy of few cm
-image acquisition sensitive to line of sight
-image granularity introduces noise, ideal resolution between 5 to 10m that capture the tree canopy seize

Difference 2022-2014

@1 m resolution

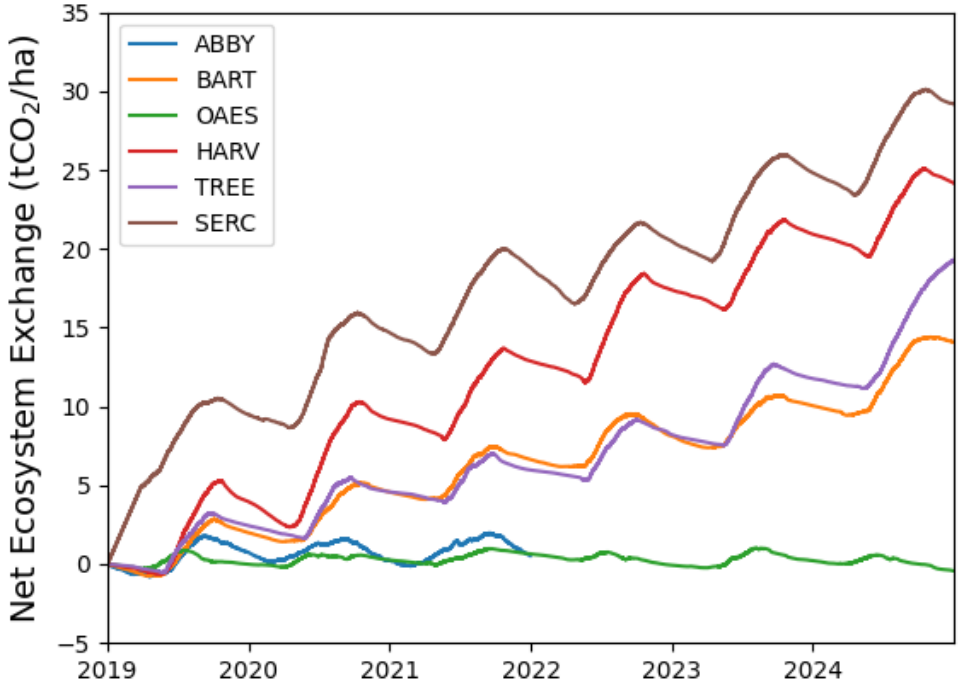


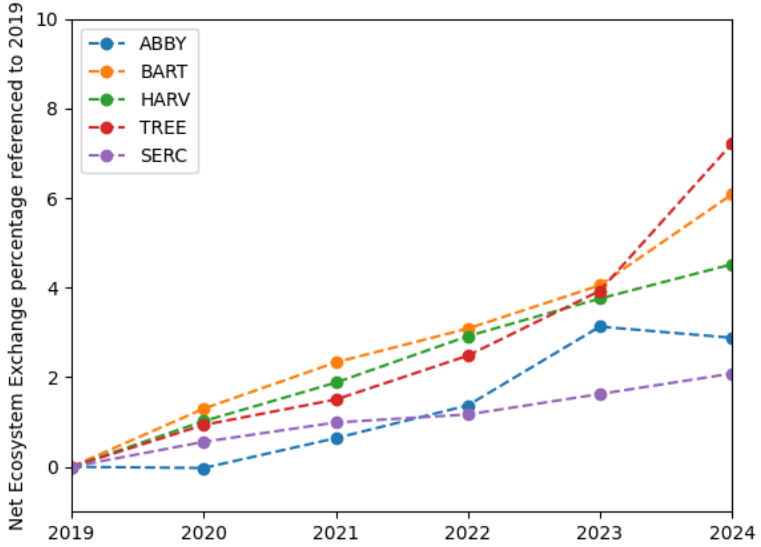
@10 m resolution



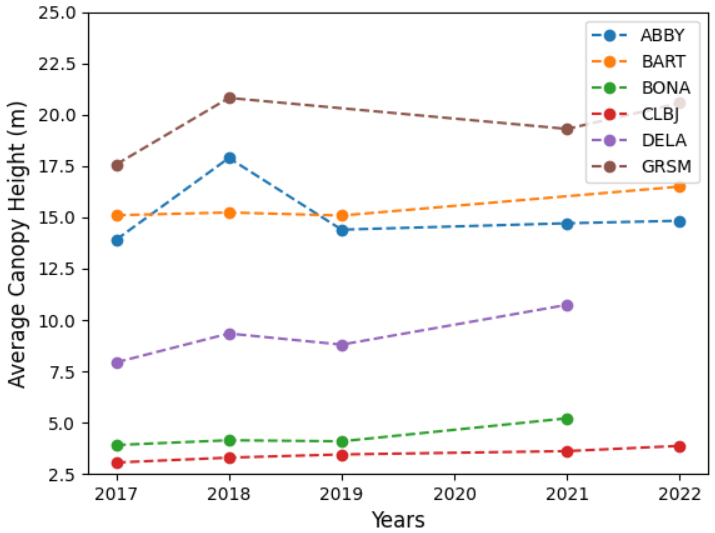
NEON Harvard site

Net Ecosystem Exchange (NEE)





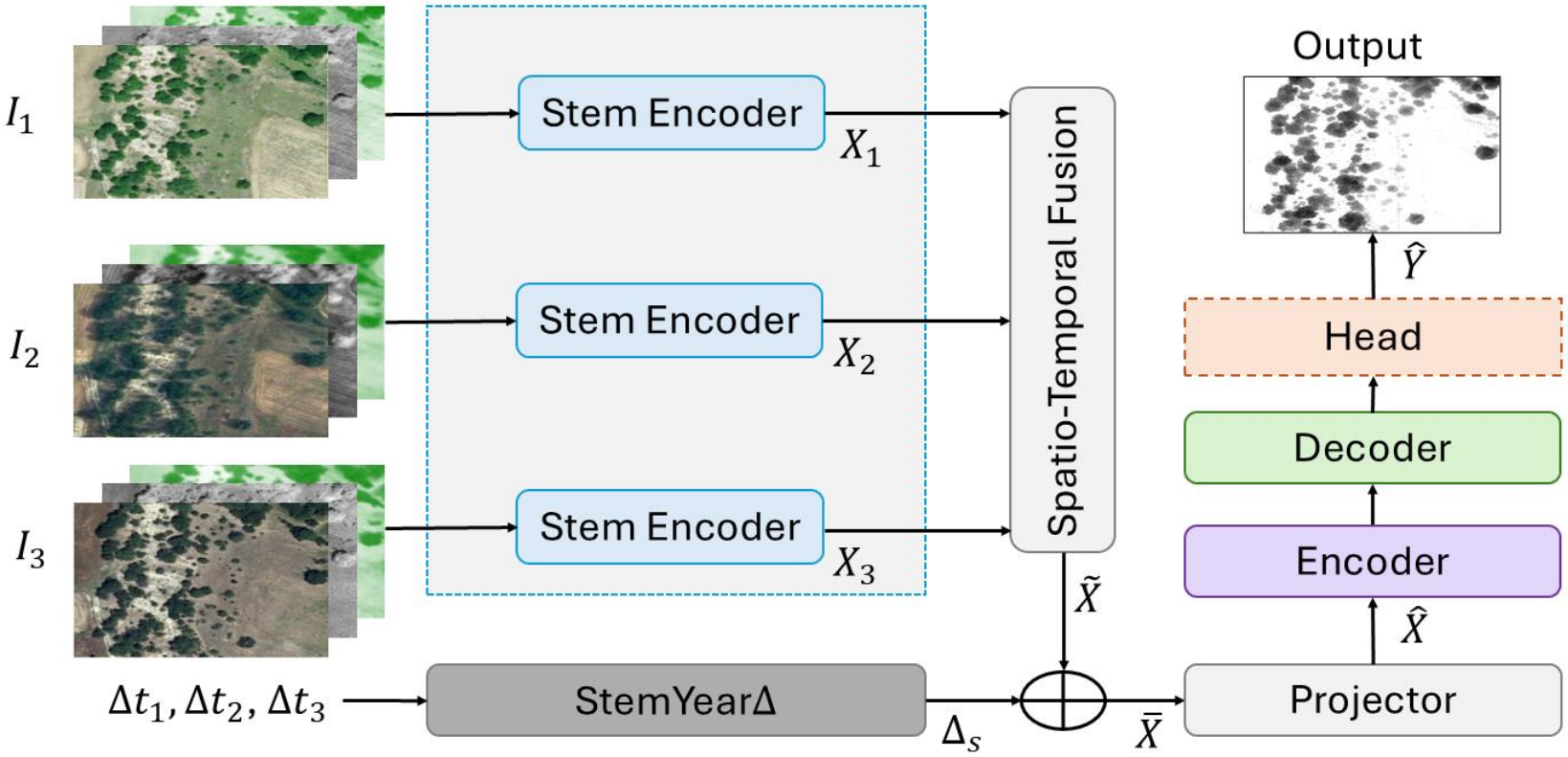
NEE growth is clean



HM growth is nosier

- NEON sites with dense forest coverage across multiple ecosystems
- Data extracted from Ameriflux QAQC (gap filled data)
- Correlate year to year change in NEE to Lidar/CHM changes

Model



Losses:

$$\mathcal{L}_{\text{wmse}}(\hat{Y}) = \frac{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left[m_{i,j} w_{i,j} (\hat{y}_{i,j} - y_{i,j})^2 \right]}{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} m_{i,j}},$$

where $w_{i,j}$ is defined as $w = 1 + k\mathbb{1}(y > \theta)$

$$\mathcal{L}_{\text{mmae}}(\hat{Y}) = \frac{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left[m'_{i,j} |\hat{y}_{i,j} - y_{i,j}| \right]}{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} m'_{i,j}},$$

where $m' = m\mathbb{1}(y > \theta)$.

Results – Architectures

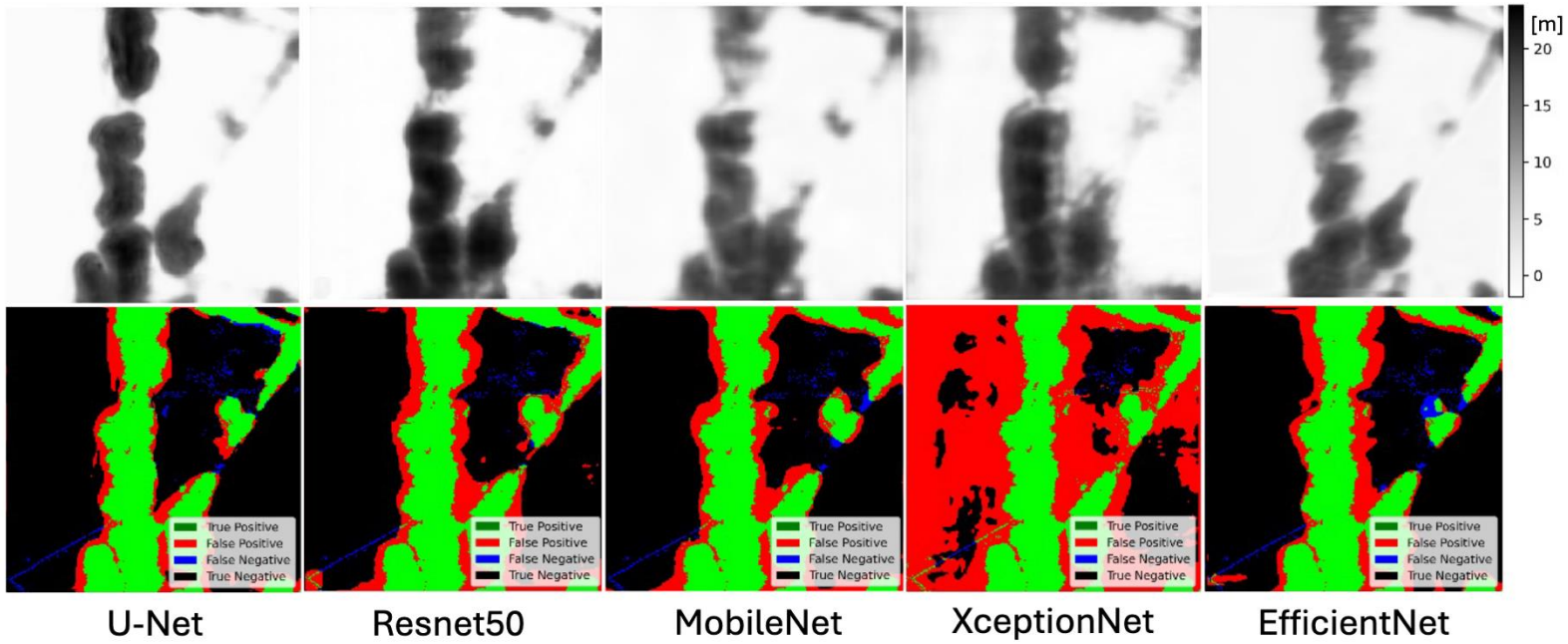
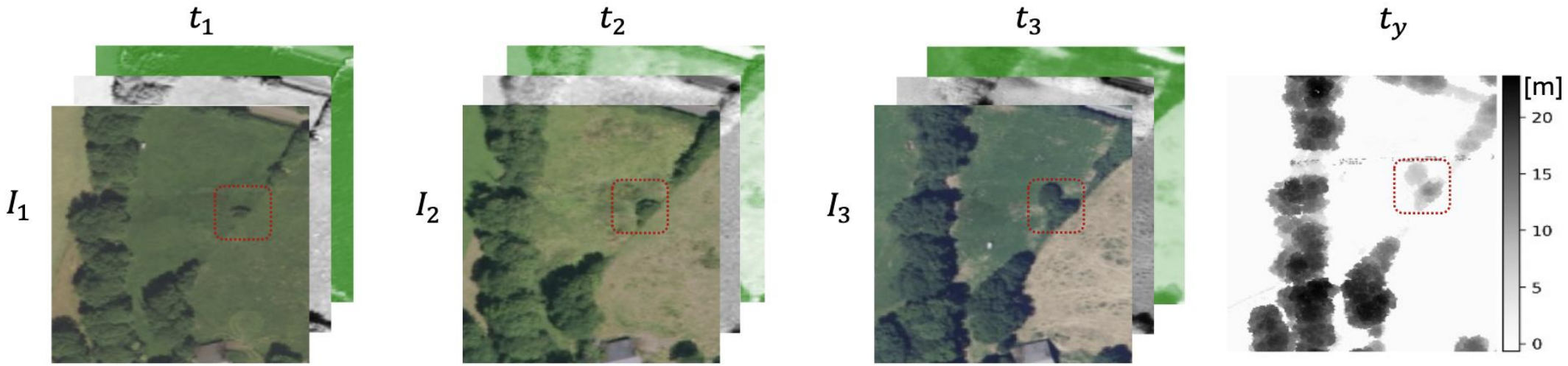


Illustration of the CHM predictions for the sample in Figure 5. In the first row, using different encoder-decoder architectures including: U-Net, Resnet50, MobileNet, XceptionNet and EfficientNet. The second row represents, w.r.t the CHM prediction and ground truth, true positives in green, false positives in red, false negatives in blue and true negatives in black.

Ex.	Encoder	Decoder	MSE	MAE
1	U-Net	U-Net	**6.47**	**11.78**
2	ResNet-50	DeepLabv3	7.37	11.95
3	MobileNet	DeepLabv3	8.26	13.34
4	XceptionNet	DeepLabv3	10.28	12.96
5	EfficientNet	DeepLabv3	8.69	13.24

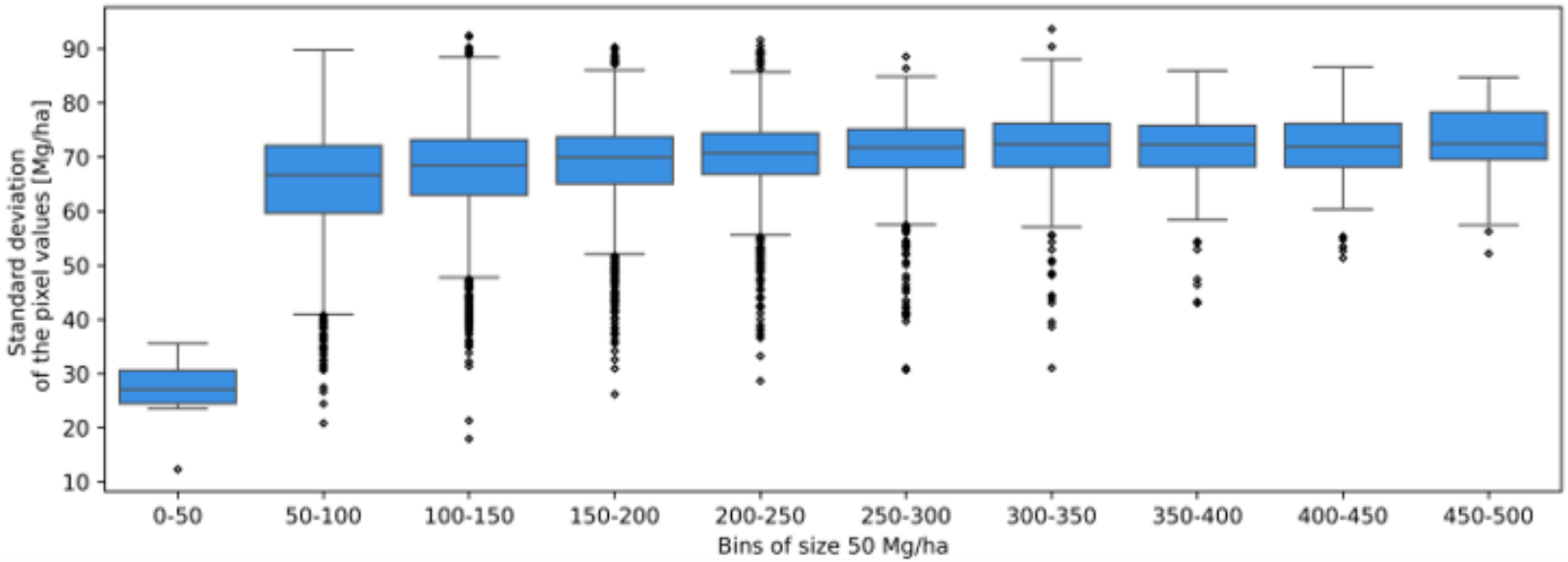
Quantitative comparison of encoder-decoder architectures. Lower is better.

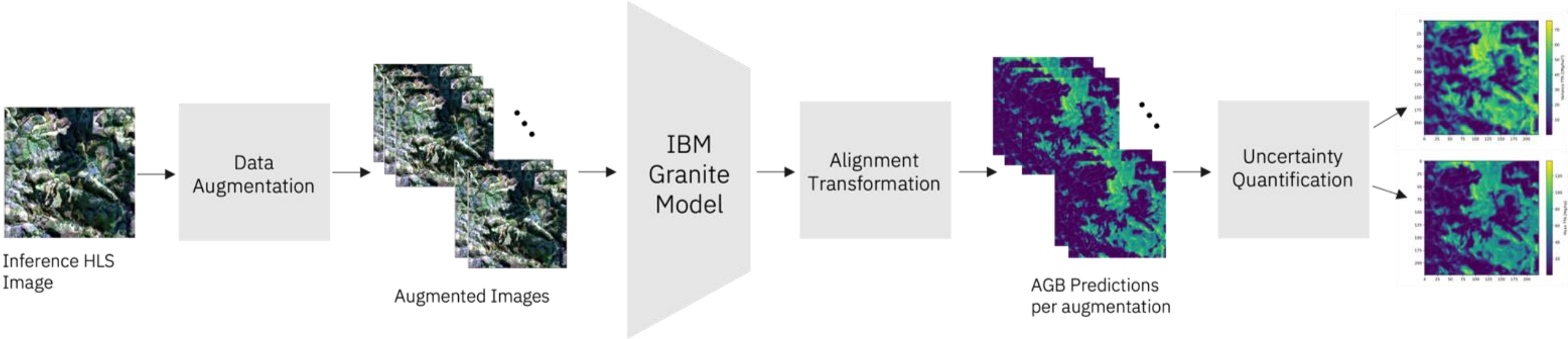
Canopy growth prediction



A sample with visible tree growth, outlined in red, from PrediTree. We collect multi-spectral (RGB+NIR+NDVI) data at three timestamps t_1, t_2, t_3 and their corresponding spatially aligned CHM tile at t_y.

Uncertainty quantification





Summary

Multimodal data can improve the prediction accuracy and reduce uncertainty of the prediction

Tree growth model improved with high fidelity NEE flux data; model considers canopy height data change

Model needs to be extended between flux tower locations and require validation (multiyear LiDAR data)

More modalities: climate, soil variables